

SHETH L.U.J. & SIR M.V. COLLEGE OF ARTS, SCIENCE & COMMERCE

Practical 9: Cleaning and Structuring Data for Analysis

AIM :

a) Identify and handle missing or inconsistent data

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Car_Sales_T...Dataset.csv

covid_19.csv

customer_purchase.csv

Department_Master.csv

Iris (1).csv

Iris (2).csv

Iris.csv

restaurant-1-orders.csv

Sem1_Scores.csv

Sem2_Scores.csv

Student_Activities.csv

students.csv

New Union

New Table

Go to Worksheet

Data Source

Sheet 1

Sem1_Scores

8 fields 10 rows

10 rows

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote File...
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name

Student ID	Full Name	Dept Info	Math	Physics	Chemistry
S101	Aarav Sharma	CSE-01	85	0.900000	88
S102	priya VERMA	ECE-02	null	0.750000	80
S103	Rohan Gupta	CSE-01	78	0.820000	null
S104	Sneha Patel	IT-03	92	0.880000	90

25°C Partly sunny

Search

ENG IN

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Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Pages

Columns

Rows

Filters

Sheet 1

Physics_Clean

[Physics] * 100

The calculation is valid.

Apply OK

Double-click or drag fields from the data pane.

Data Source

Sheet 1

25°C Partly sunny

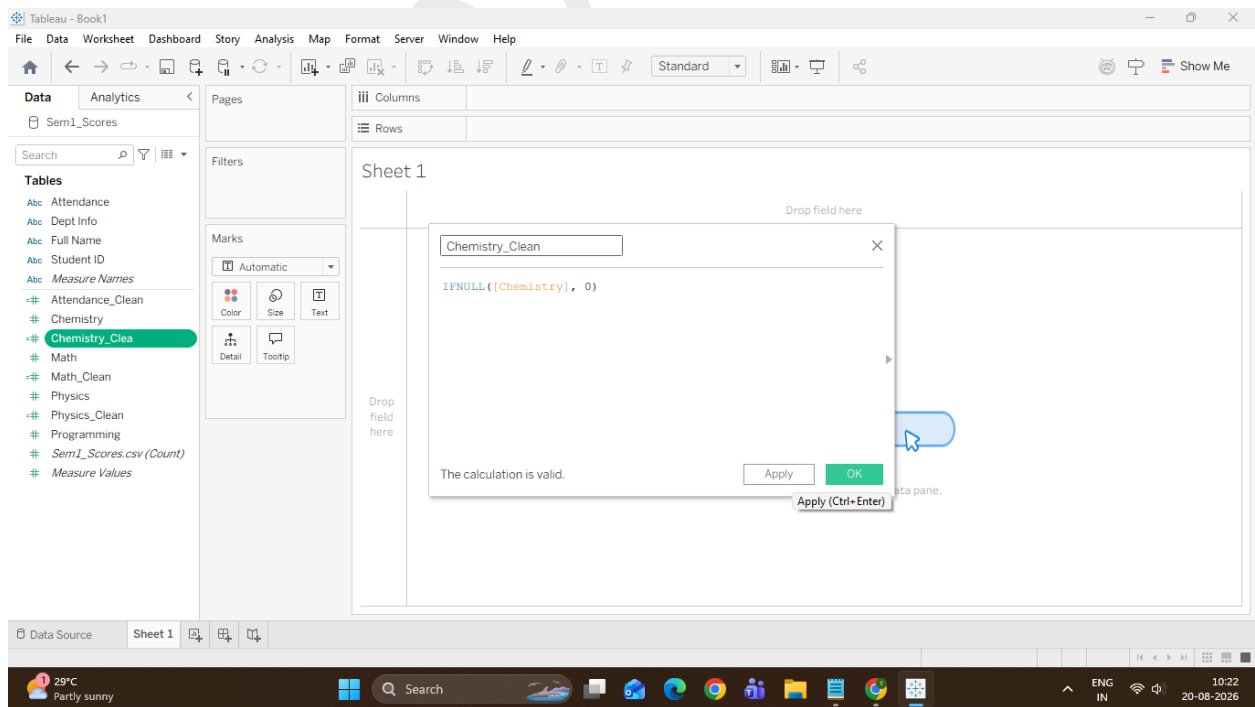
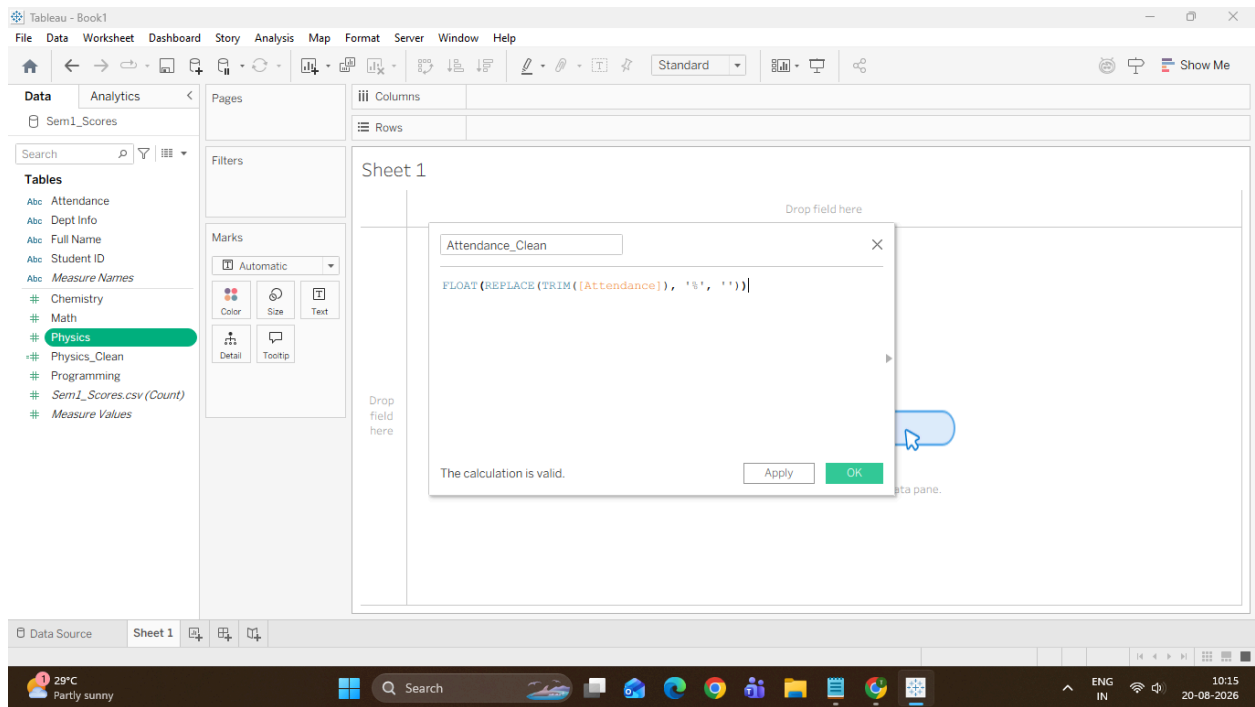
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Practical 9: Cleaning and Structuring Data for Analysis



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Practical 9: Cleaning and Structuring Data for Analysis

b) Convert wide data into tall format

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Car_Sales_T...Dataset.csv
covid_19.csv
customer_purchase.csv
Department_Master.csv
Iris (1).csv
Iris (2).csv
Iris.csv
restaurant-1-orders.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv
students.csv
New Union
New Table Extension

Sem1_Scores

Connection: Live

Filters: 0 | Add

Sem1_Scores.csv

10 fields 40 rows

40 rows

Name: Sem1_Scores.csv

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name
Abc	Dept Info	Sem1_Scores.csv	Dept_Info
Abc	Attendance	Sem1_Scores.csv	Attendance
Abc	Subject	Pivot	Pivot Field Na...
#	Score	Pivot	Pivot Field Val...
#	Physics_Cle	Calculation	Calculation_02...
#	Attendance_C	Calculation	Calculation_02...
#	Math_Clear	Calculation	Calculation_02...
#	Chemistry_I	Calculation	Calculation_02...

Dept Info	Attendance	Subject	Score	Physics_Cle
CSE-01	85 %	Chemistry	88.0000	null
CSE-01	85 %	Math	85.0000	null
CSE-01	85 %	Physics	0.9000	null
CSE-01	85 %	Programming	95.0000	null
ECE-02	90 %	Chemistry	80.0000	null
ECE-02	90 %	Math	null	null
ECE-02	90 %	Physics	0.7500	null
ECE-02	90 %	Programming	85.0000	null
CSE-01	78 %	Chemistry	null	null
CSE-01	78 %	Math	78.0000	null
CSE-01	78 %	Physics	0.8200	null
CSE-01	78 %	Programming	88.0000	null
IT-03	95 %	Chemistry	90.0000	null

c) Perform union of multiple files

Tableau - Book1

File Data Server Window Help

Connections

Sem2_Scores
Text file

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Car_Sales_T...Dataset.csv
covid_19.csv
customer_purchase.csv
Department_Master.csv
Iris (1).csv
Iris (2).csv
Iris.csv
restaurant-1-orders.csv
Sem1_Scores.csv
Sem2_Scores.csv
New Union
New Table Extension

Sem2_Scores.csv (Multiple Connections)

Connection: Live

Filters: 0 | Add

Sem2_Scores.csv

10 rows

Name: Sem2_Scores.csv

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student ID	Sem2_Scores.csv	Student_ID
Abc	Full Name	Sem2_Scores.csv	Full_Name
Abc	Dept Info	Sem2_Scores.csv	Dept_Info
#	Math	Sem2_Scores.csv	Math
#	Physics	Sem2_Scores.csv	Physics
#	Chemistry	Sem2_Scores.csv	Chemistry
#	Programming	Sem2_Scores.csv	Programming
Abc	Attendance	Sem2_Scores.csv	Attendance

Math	Physics	Chemistry
88	0.920000	90
79	0.780000	82
82	0.850000	80
95	0.900000	94
52	0.580000	55
90	0.940000	88
74	0.730000	78
96	0.960000	95
70	0.650000	62
85	0.880000	84

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The screenshot shows the Tableau Desktop interface. The 'Connections' pane on the left lists 'Sem1_Scores' as a text file. The 'Files' pane shows various CSV files, including 'Sem1_Scores.csv'. The main workspace displays a 'Union' operation between 'Sem1_Scores' and 'Department_Master'. The 'Fields' shelf contains 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', and 'Chemistry'. The data preview shows 20 rows of data.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry
S107	Rahul Nair	ECE-02	74	0.730000	78
S108	Kavya Reddy	IT-03	96	0.960000	95
S109	Vikram Joshi	CSE-01	70	0.650000	62
S110	Pooja Iyer	ECE-02	85	0.880000	84

d) Execute cross-database joins

The screenshot shows the Tableau Desktop interface with an error message: 'Union is made of multiple tables.' The error message indicates that the 'Union' operation is not supported for cross-database joins. The 'Fields' shelf contains 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', 'Programming', and 'Student_ID'. The data preview is unavailable.

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Practical 9: Cleaning and Structuring Data for Analysis

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Car_Sales_T...Dataset.csv
covid_19.csv
customer_purchase.csv
Department_Master.csv
Iris (1).csv
Iris (2).csv
Iris.csv
restaurant-1-orders.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv
students.csv

New Union
New Table Extension

Union is made of multiple tables. 0

Union

Join

Inner Left Right Full Outer

Data Source

Dept_Info = Dept_Master.csv

Search

Department Name
Dept Code
Dept Target Ave
HOD Name
Total Dept Capacity
Create Join Calculation...

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name

Data Source

Sheet 1

30°C
Partly sunny

Search

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20-08-2026

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Car_Sales_T...Dataset.csv
covid_19.csv
customer_purchase.csv
Department_Master.csv
Iris (1).csv
Iris (2).csv
Iris.csv
restaurant-1-orders.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv
students.csv

New Union
New Table Extension

Union is made of multiple tables. 0

Union

Join

Inner Left Right Full Outer

Data Source

Dept_Info = Dept_Master.csv

Search

Department Name
Dept Code
Dept Target Ave
HOD Name
Total Dept Capacity
Create Join Calculation...

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name

Data Source

Sheet 1

30°C
Partly sunny

Search

ENG IN

10:57
20-08-2026

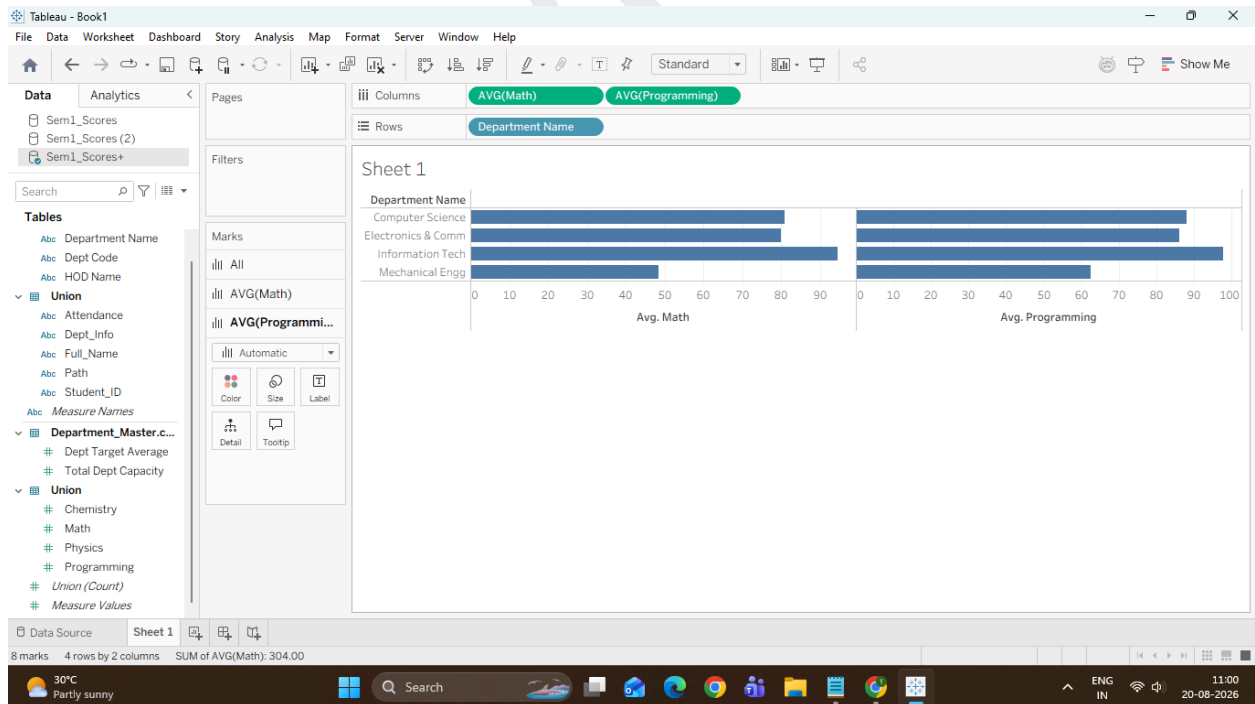
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Practical 9: Cleaning and Structuring Data for Analysis

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Sem1_Scores' as a text file. Below it, the 'Files' pane shows a list of CSV files including 'Car_Sales_T...Dataset.csv', 'covid_19.csv', 'customer_purchase.csv', 'Department_Master.csv', 'Iris (1).csv', 'Iris (2).csv', 'Iris.csv', 'restaurant-1-orders.csv', 'Sem1_Scores.csv', 'Sem2_Scores.csv', 'Student_Activities.csv', and 'students.csv'. The main workspace displays a 'Union' of two data sources: 'Sem1_Scores' and 'Department_Master.csv'. A tooltip explains that a union is made of multiple tables. Below this, a data preview table shows the combined data with columns: Student_ID, Full_Name, Dept_Info, Math, Physics, and Chemistry. The bottom status bar indicates 'Data Source', 'Sheet 1', and '20 rows'.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry
S101	Aarav Sharma	CSE-01	85	0.900000	88
S102	priya VERMA	ECE-02	null	0.750000	80
S103	Rohan Gupta	CSE-01	78	0.820000	null
S104	Sneha Patel	IT-03	92	0.880000	90

e) Restructure data for better visualization



f) Handle different levels of detail in datasets

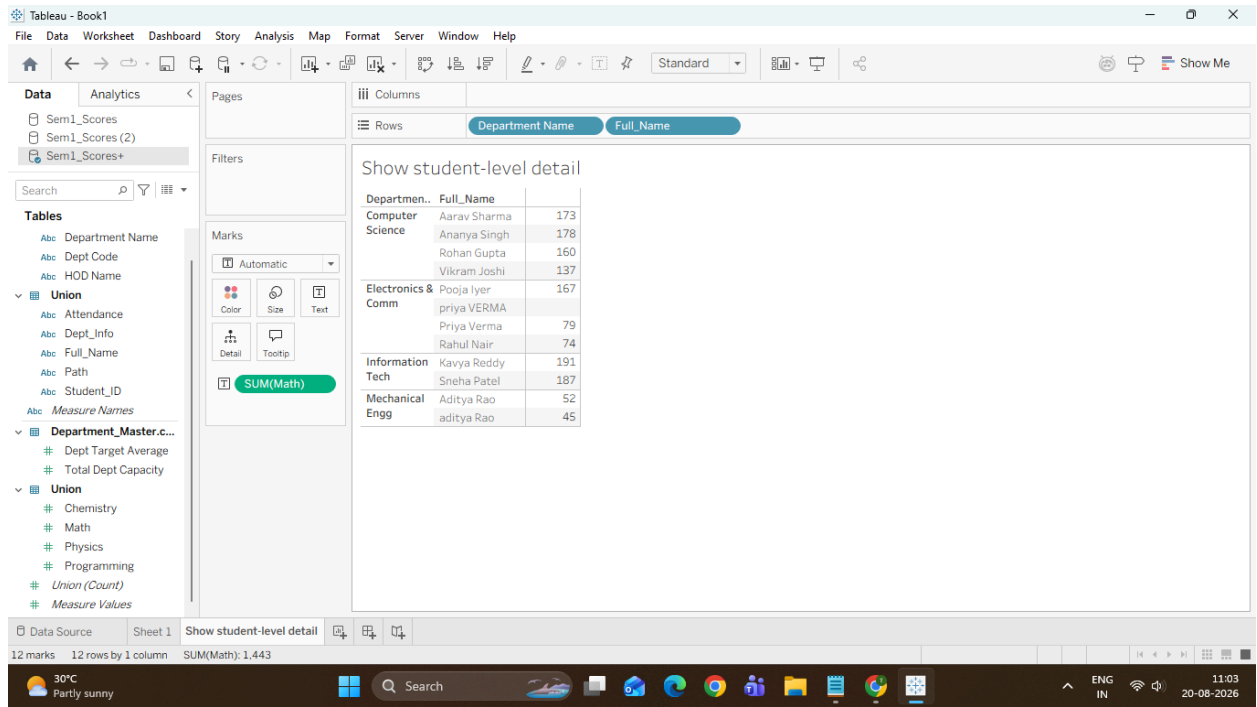


Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Back Forward Refresh Download Copy Paste Undo Redo Erase Wipe Clean Wipe All Wipe All & Refresh

Standard

Show Me

Data Analytics

Sem1_Scores
Sem1_Scores (2)
Sem1_Scores+

Search

Tables

- Department Name
- Dept Code
- HOD Name
- Union
 - Attendance
 - Dept_Info
 - Full_Name
 - Path
 - Student_ID
 - Measure Names
- Department_Master.c...
 - Dept Target Average
 - Total Dept Capacity
- Union
 - Chemistry
 - Math
 - Physics
 - Programming
 - Union (Count)
 - Measure Values

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

SUM(Math)

Columns

Rows

Department Name Full Name

Show student-level detail

Department Name	Full Name	Score
Computer Science	Aarav Sharma	173
Computer Science	Ananya Singh	178
Computer Science	Rohan Gupta	160
Computer Science	Vikram Joshi	137
Electronics & Comm	Pooja Iyer	167
Electronics & Comm	priya VERMA	79
Electronics & Comm	Priya Verma	74
Electronics & Comm	Rahul Nair	74
Information Tech	Kavya Reddy	191
Information Tech	Sneha Patel	187
Mechanical Engg	Aditya Rao	52
Mechanical Engg	aditya Rao	45

Data Source Sheet 1 Show student-level detail

12 marks 12 rows by 1 column SUM(Math): 1,443

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Search

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g) Apply advanced data cleaning techniques

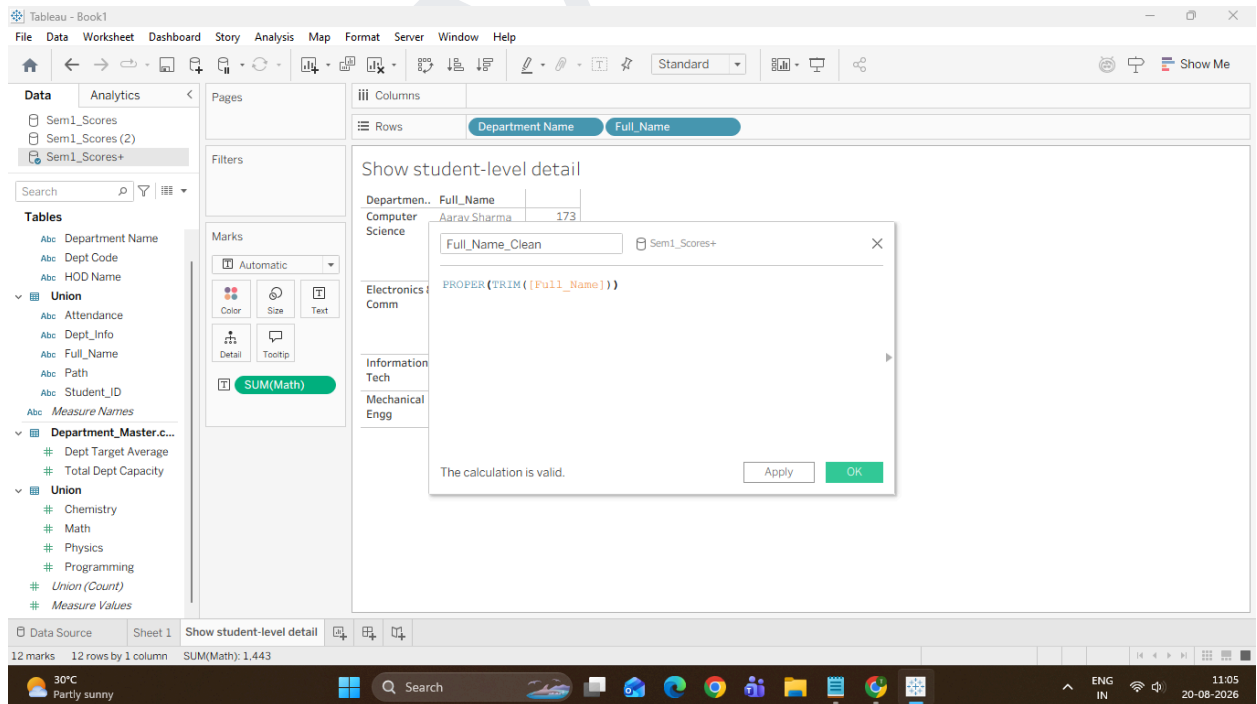


Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Back Forward Refresh Download Copy Paste Undo Redo Erase Wipe Clean Wipe All Wipe All & Refresh

Standard

Show Me

Data Analytics

Sem1_Scores
Sem1_Scores (2)
Sem1_Scores+

Search

Tables

- Department Name
- Dept Code
- HOD Name
- Union
 - Attendance
 - Dept_Info
 - Full_Name
 - Path
 - Student_ID
 - Measure Names
- Department_Master.c...
 - Dept Target Average
 - Total Dept Capacity
- Union
 - Chemistry
 - Math
 - Physics
 - Programming
 - Union (Count)
 - Measure Values

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

SUM(Math)

Columns

Rows

Department Name Full Name

Show student-level detail

Department Name	Full Name	Score
Computer Science	Aarav Sharma	173

Full_Name_Clean

Sem1_Scores+

PROPER(TRIM([Full_Name]))

The calculation is valid.

Apply OK

Data Source Sheet 1 Show student-level detail

12 marks 12 rows by 1 column SUM(Math): 1,443

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Search

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